

William Farlessyost

Lafayette, IN | (828) 319-5078 | wbfarlessyost@gmail.com
farlessyost.github.io | linkedin.com/in/farlessyost

SUMMARY

Mechatronics engineer with a Ph.D. and experience in controls, system identification, dynamic modeling, robotics, and machine-learning-assisted engineering. Develops physical-system models, control software, embedded firmware, and mechanical designs, including current medical-scanner positioning and camera-free caregiver sensing prototypes.

TECHNICAL SKILLS

Controls & Modeling: System identification, nonlinear dynamics, reduced-order and mechanistic modeling, inverse kinematics, sensor fusion, model validation, MATLAB/Simulink, ASPEN Plus/Dynamics

Robotics & Embedded Systems: ROS / ROS 2, MuJoCo, Gazebo, MoveIt!, SLAM, motion planning, ESP32-C3, ESP-NOW, motor-driver interfaces, sensor integration

Programming & Data: C/C++, Python, MATLAB, R, SQL/SQLite, Bash, Git; TensorFlow, PyTorch, Scikit-learn, SINDy, liquid time-constant neural networks

Mechanical & Engineering Tools: FreeCAD, SolidWorks, AutoCAD, 3D Slicer, BioCro, SimaPro

CURRENT ENGINEERING PROJECTS

TetherXYZ | Cable-Driven Medical-Scanner Positioning

Independent engineering project | In development

- Designed a four-cable positioning mechanism and two-axis tool wrist in FreeCAD, with modular winches, printable parts, and a frame intended for local fabrication and repair.
- Developed pulley-aware inverse kinematics, homing and surface-tracking simulations, and a MuJoCo model with ROS 2 Cartesian commands.
- Implemented ESP32-C3 motor-control firmware with acceleration limits, arming logic, a command watchdog, and CRC-protected communication between controller boards.
- Modeled stereo-camera and laser-line surface reconstruction and evaluated assembly fit, cable geometry, and tool trajectories for prototype development.

Peepin on Papaw | Camera-Free Home Activity Sensing

Independent engineering project | In development

- Developed ESP32-C3 motion and magnetic-door sensor firmware and an ESP-NOW receiver with heartbeats, debouncing, duplicate suppression, and OLED status feedback.
- Built Python/SQLite event logging and a browser-based caregiver dashboard; verified live sensor packets from three physical locations.
- Implemented graph-based room estimation with explicit uncertainty and scenario tests for packet loss, repeated triggers, visitors, and disconnected sensors.
- Designed sensor and receiver enclosures in FreeCAD, integrating electronics, battery packaging, and printable mechanical features.

ENGINEERING & RESEARCH EXPERIENCE

Graduate Research Assistant

Aug. 2020 - May 2025

Purdue University | Sustainable Industrial-Natural Coupled Systems Group

- Applied nonlinear system identification to chemical-process and biological-system data to derive compact, interpretable dynamic models.
- Combined machine learning with mechanistic simulation and developed surrogate models for coupled natural-industrial systems under changing operating and climate conditions.
- Analyzed robustness, sensitivity, observability, and reduced-sensor configurations using Python and MATLAB; communicated findings through publications and conference presentations.

ENGINEERING & RESEARCH EXPERIENCE (CONTINUED)

Undergraduate Research Assistant, Controls

Aug. 2019 - Jan. 2020

University of North Carolina Asheville | Asheville, NC

- Validated and implemented a quadrotor UAV control system for tracking and softly landing on a moving platform, integrating fiducial recognition, sensor fusion, ROS, and Simulink in simulation and physical tests.

Mechatronics Senior Design Project

Aug. 2019 - May 2020

NASA Lunabotics Challenge | Asheville, NC

- Designed and built two autonomous double-Ackermann-steered mining robots; contributed to mechanical and electrical integration, SLAM, path planning, navigation, and closed-loop control.

Undergraduate Research Assistant, Robotics

Aug. 2019 - May 2020

University of North Carolina Asheville | NSF-Funded Research

- Developed an apprenticeship-learning system for a UR3e collaborative robot with an RG2 gripper to sort previously unseen objects; implemented motion planning and simulation using ROS, MoveIt!, and Gazebo.

Undergraduate Research Assistant, Robotics

Sept. 2017 - May 2019

University of North Carolina Asheville | Asheville, NC

- Developed a Jenga-playing robot for reinforcement-learning research, including mechanical and electrical design, ROS integration, motion planning, and Gazebo simulation.

ADDITIONAL PROFESSIONAL EXPERIENCE

Founder & AI Product Developer

May 2025 - Present

Fee Faux Fum | Remote

- Built a live beta product using Python, FastAPI, Supabase, document parsing, structured extraction, rule-based analysis, and automated reporting; managed architecture, debugging, deployment, and iterative validation.

AI Trainer, Multimodal AI Evaluation

Dec. 2025 - Present

Handshake AI Fellowship | Remote

- Evaluated technical outputs for correctness, reasoning quality, consistency, and instruction following using structured assessment criteria.

EDUCATION

Purdue University

2025

Ph.D., Agricultural and Biological Engineering

Majors: Computational Science & Engineering; Ecological Sciences & Engineering

University of North Carolina Asheville

2020

B.S.E., Mechatronics Engineering (joint NCSU-UNCA program), *summa cum laude*

B.A., Applied Mathematics, *summa cum laude*

SELECTED PUBLICATIONS & AWARDS

Farlessyost W, Singh S. "Reduced order dynamical models for complex dynamics in manufacturing and natural systems using machine learning." *Nonlinear Dynamics*. 2022; 110(2):1613-1631.

Farlessyost W, Singh S. "Improving Mechanistic Model Accuracy with Machine Learning Informed Physics." *Foundations of Computer Aided Process Design*, 2024.

NSF Graduate Research Fellowship Program, 2022-2025.

Departmental Professionalism and Academic Excellence Award and Top Math Student Award, UNCA, 2020; SIMIODE Differential Equations Modeling Challenge, Meritorious Award, 2019.

TEACHING

Graduate and Undergraduate Teaching Assistant | Jan. 2017 - May 2021. Led laboratories, help sessions, grading, and project support in automatic control, computer systems, thermodynamics, circuits, statics, and statistical modeling.